MAT 362 HW10

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Problem. 4.4.7. Intersect the cylinder $x^2 + y^2 = 1$ with a plane passing through the x-axis and making an angle of θ with the xy-plane.

(a) Show that the intersecting curve is an ellipse C.

(b) Compute the absolute value of the geodesic curvature of C in the cylinder at the points where C meets its axes.

Solution. (a) Consider the basis of the plane given by $v_1 = (1, 0, 0)$ and the orthogonal vector $v_2 = (0, \cos\theta, \sin\theta)$. Given a point (x, y, z) on C, we have $(x, y, z) = xv_1 + \frac{y}{\cos\theta}v_2$ (since $y/\cos\theta$ is the only choice of coefficient that matches with the second component of (x, y, z)). We have that $x^2 + y^2 = 1$ a priori, so, within the intersecting plane, C has equation $(x')^2 + (y')^2\cos^2\theta = 1$, where points (x', y') are respectively the v_1, v_2-coordinates with respect to the orthonormal basis $\{v_1, v_2\}$ of the plane; this is exactly the equation of an ellipse.

(b) At a given point p, since $k^2 = k_g^2 + k_n^2$, we have that $|k_g| = k\sin\phi$ where $\phi \in [0, \pi)$ is the angle between the normal to the curve and normal to the surface at p.

The axes of C intersect C at two points on the x-axis, where $\phi = 0$ (hence the geodesic curvature is 0), and two points on the yz-plane, where $\phi = \theta$. In the latter case we wish to compute the curvature of the ellipse at $(0, 1, \tan\theta)$ and $(0, -1, -\tan\theta)$; since both values are identical by symmetry we just compute for the former. We can parametrize the ellipse as $r(t) = (\cos t, \sin t/\cos\theta, 0)$ (with respect to orthonormal basis v_1, v_2, $v_1 \times v_2$) so that the curvature at $t = \frac{\pi}{2}$ is

$$\frac{|r' \times r''|}{|r'|^3} = \frac{|(0, 0, 1/\cos\theta)|}{|(-1, 0, 0)^3|} = \sec\theta.$$

The geodesic curvature is thus $\tan\theta$. □

Problem. 4.4.11. State precisely and prove: the algebraic value of the covariant derivative is invariant under orientation-preserving isometries.

Solution. Statement: Let w be a differentiable field of unit vectors along a parametrized curve $\alpha\colon I \to S$ on an oriented surface S. Then, if f is an isometry of S, the algebraic value of $f \circ w$ (which is a vector field on $f \circ \alpha$) is equal to the algebraic value of w. We thus want to show that if

$$\frac{Dw}{dt} = \lambda(N \wedge w(t)), \quad \frac{D(f \circ w)}{dt} = \lambda'(f(N) \wedge f(w(t))$$

then $\lambda = \lambda'$. Being an isometry, $\langle N, w(t)\rangle = \langle f(N), f(w(t))\rangle$, as well as $|N| = |f(N)|$ and $|w(t)| = |f(w(t))|$, so $|N \wedge w(t)| = |f(N) \wedge f(w(t))$; thus it suffices to show that the magnitudes of Dw/dt and $D(f \circ w)/dt$ are the same.

On the other hand Dw/dt is just the projection of w' on $T_{\alpha(0)}(S)$ and $D(f \circ w)/dt$ is the projection of $(f \circ w)'$ on $T_{f(\alpha(0))}(S)$, but f being an isometry means the latter is the same length as the former ("the geometric configurations of the objects are the same" roughly), which gives us the result. □

Problem. 4.4.18. Consider a geodesic which starts at a point p in the upper part of a hyperboloid of revolution $x^2 + y^2 - z^2 = 1$ and makes an angle θ with the parallel passing through p in such a way that $\cos\theta = 1/r$, where r is the distance from p to the z-axis. Show that by following the geodesic in the direction of decreasing parallels, it approaches asymptotically the parallel $x^2 + y^2 = 1$, $z = 0$.

Solution. Parametrize the geodesic by arc length starting at p to get the map $\alpha\colon [0,\infty) \to H$ where H is the hyperboloid. Then let $r(s)$ be the distance to the z axis and $\theta(s)$ the angle between $\alpha'(s)$ and the parallel passing through $\alpha(s)$. Then,

$$r(s)\cos\theta(s) = r(0)\cos\theta(0) = r \cdot 1/r = 1$$

for all s. Now let $c = \inf_{s\geq 0} r(s) = \lim_{s\to\infty} r(s)$. We know that $\sqrt{1-\frac{1}{c}} < \sin\theta(s)$ for all s and, if $\alpha(s) = (x(s), y(s), z(s))$, then $z'(s) = \sin\theta(s)$ because $|\alpha'(s)| = 1$. Thus if $c > 1$, given $\varepsilon > 0$, pick t_ε so that $r(t_\varepsilon) < c + \varepsilon$. Because of the bound on z', $z(t+1) \leq z(t) - \varepsilon$. Clearly $z > 0$ for all s (otherwise we cross through the xy-plane, when $r(s) = 1 < c$), but this is a contradiction, since we can just take n large enough so that $z(t+n) \leq z(t) - n\varepsilon < 0$.

That is, $c > 1$ leads to a contradiction, so the limiting value of r must be 1 (since of course $r(s) \geq 1$) and the limiting value of $\cos\theta(s)$ must also be 1 because of Clairaut's relation, which is what we want. □

Problem. 4.4.23. Show that the isometries of the unit sphere S^2 are the restrictions to S^2 of the linear orthogonal transformations of $\mathbb{R}^3$.

Solution. Let f be an isometry of S^2. Then f maps great circles to great circles (since it maps geodesics to geodesics, and the great circles are all the geodesics of S^2) and preserves the "angular distance" between two points on S^2 (i.e. $\langle p, q\rangle = \langle f(p), f(q)\rangle$). Extending this map to $f(tp) = tf(p)$ for $p \in S^2$ and real t, we can write, for arbitrary $t_1p_1, t_2p_2 \in \mathbb{R}^3$ (with $p_1, p_2 \in S^2$), that $\langle t_1p_2, t_2p_2\rangle = \langle t_1 f(p_1), t_2 f(p_2)\rangle$, i.e. f preserves inner products (because distances and angles are preserved), which is enough to conclude that it is an orthogonal linear transformation by a previous homework problem. □